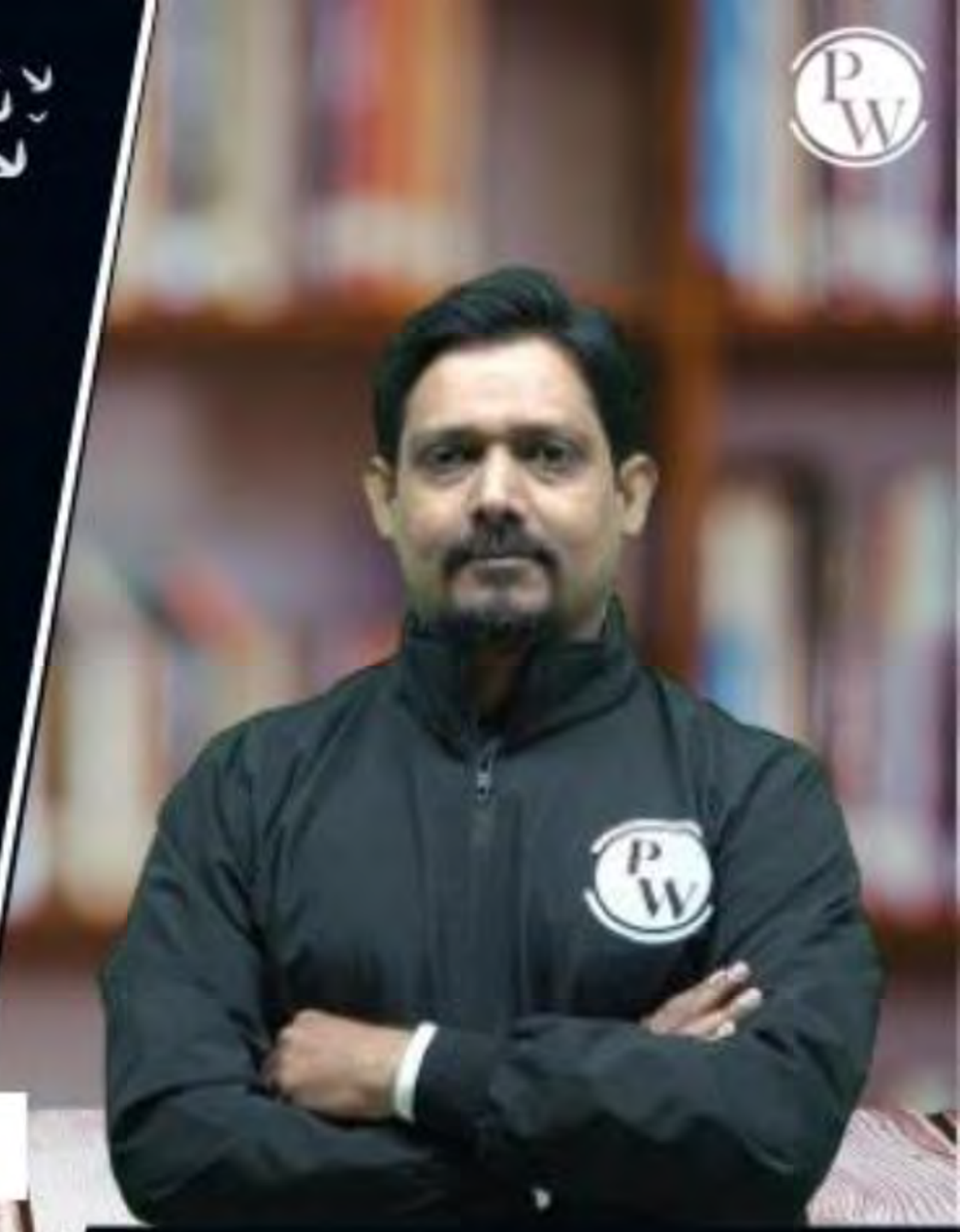
Computer Science & IT



Data Structure & Programming

Array

Lecture No. 03



By- Abhishek Sir

Recap of Previous Lecture



- Topic: 1-D Array ✓
- Topic: 2-D Array ✓
- Topic: Row Major Order ✓
- Topic
- Topic

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Topics to be Covered



Topic 2-D Array Column Major Order

Topic Lower Triangular Matrix

Topic Upper Triangular Matrix

} Sparse Matrix

Topic

Topic



Question : 1-D Array

#Q

5 mark 1892

Consider the above array A[-5.....100]. Base address of the array is 1000 and each element occupies 2-Bytes of space. What is the address of A[70]?

(A) 1150

(B) 1250

(C) 1200

(D) 1350

LB

UB

* easy

difficult to handle

$$BA + (i - LB) \times Size$$

$$1000 + (70 - (-5)) \times 2 = 1000 + 75 \times 2$$

$$1000 + 150 = 1150$$

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Question : 2-D Array, MSQ

#Q

[A][B]

Consider the following array A[-23...93] [-7...31].Base address is 1000 and each element occupies 2-Bytes of space. Which of the following is TRUE ?

(A) The number of rows is 117 elements in a row

(B) The Array B[0..116] [2..40] have same number of elements as the array declared in question

(C) Address of A[0][5] is 3538 row major order

(D) Address of A[0][0] is 3538 column major order

31 - (-7) + 1

31 + 7 + 1 = 39

UB - LB + 1 = 93 - (-23) + 1

40 - 2 + 1 = 39

117

Ignore

B[0...116] = 116 - 0 + 1 = 117

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Question : 2-D Array, MSQ

Consider the following array A[-23...93] [-7...31].Base address is 1000 and each element occupies 2-Bytes of space. Which of the following is TRUE ?

(C) Address of A[0][5] is 3538 row major order



0th Row (-23) — (-1)
LB UB
-1-(-23)+1 = 23

(i - L6)

(0-(-23)) = 23

* No. of rows arranged : -23 ... -1

No. of element in each row : 39

total = 23x39

If 5th row we are in 5th column

(5-(-7)) = 12 [(j - L6)]

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$$1000+(23\times39+12)\times2=2818$$

$$1000+(\quad 909 \quad)\times2$$

$$(i-(\quad$$

$$1000+[(0-(-23))\times(31-(-7)+1)+(5-(-7)]\times2$$

using formula

$$1000+[23\times39+12]\times2$$



Question : 2-D Array, MSQ

Consider the following array A[-23…93] [-7…31].Base address is 1000 and each element occupies 2-Bytes of space. Which of the following is TRUE ?

(D) Address of A[0][0] is 3538 column major order

A[0_116][2_40]

Rows : 116-0+1 = 117

Column = 40-2+1 = 39

UB - LB + 1

Rows 93 - (-23) + 1 = 117 ✓

Column : 31 - (-7) + 1 = 39

Slide

Topic : 2-D Generalization Row major order:

$A[Lb_1 .. ub_1][Lb_2 ... ub_2]$

Row Column

$Lb_1 \ldots (i-1)$

$i - 1 - Lb_1 + 1$



Address $A[i][j]$

$$BA + \left[(i - Lb_1) \times (ub_2 - Lb_2 + 1) + (j - Lb_2) \right] \times Size$$

In last Row No. of columns completed

No. of Raws Complete

No. of element in each Raw

$Lb_2 \longrightarrow j - 1 = j - 1 - Lb_2 + 1$

Slide



Topic : 2-D Column Major Order

BA = 1000

Size = 2B

	Column-1	Column-2	Column-3
Row-1	A[1][1]	A[1][2]	A[1][3]
Row-2	A[2][1]	A[2][2]	A[2][3]
Row-3	A[3][1]	A[3][2]	A[3][3]

A[1..3][1...3]

2D mapped as 1D array

1000	1002	1004	1006	1008	1010	1012	1014	1016
A[1,1]	A[2,1]	A[3,1]	A[1,2]	A[2,2]	A[3,2]	A[1,3]	A[2,3]	A[3,3]

Column-1 Column-2 Column-3

Slide

Topic : 2-D Column Major Order

BA = 1000
Size = 2B

	Column-1	Column-2	Column-3
Row-1	A[1][1]	A[1][2]	A[1][3]
Row-2	A[2][1]	A[2][2]	A[2][3]
Row-3	A[3][1]	A[3][2]	A[3][3]

2D mapped as 1D array

1000	1002	1004	1006	1008	1010	1012	1014	1016
A[1,1]	A[2,1]	A[3,1]	A[1,2]	A[2,2]	A[3,2]	A[1,3]	A[2,3]	A[3,3]

A[1..3][1...3]

A[2][3]

No of column Completed = 2

No. of element in each column : 3

total = 2 × 3 = 6

In 3rd column No. of Rows compled : (2-1) = 1

total : 6 + 1 = 7

Address of A[2][3] = 1000 + 7 × 2 = 1014

(3-1)

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Topic : 2-D Column Major Order

$A[Lb_1 \ldots Ub_1][Lb_2 \ldots Ub_2]$

Address $A[i][j]$

$$BA + [(j - Lb_2) \times (Ub_1 - Lb_1 + 1) + (i - Lb_1)] \times Size$$

Column major orders.

Slide



Topic : 2-D Column Major Order

#Q

Consider the following array A[-4...100] [3...100]. Base address is 1000 and each element occupies 2-Bytes of space. What is the address of A[50][48] in column major order ?

A[-4..100] [3..100]

$$1000 + [(48-3) \times (100-(-4)+1) + (50-(-4))] \times 2$$

(i - L6,)

$$1000 + [45 \times 105 + 54] \times 2 = 10558$$

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Topic : 2-D Column Major Order

Consider the following array A[-4...100][3...100]. Base address is 1000 and each element occupies 2-Bytes of space. What is the address of A[50][48] in column major order ?

$1000 + (49 \times 105 + 54) \times 2$

$= 10558$

In 48th column No. of Rows completed

$-4 - 49$

$49 - (-4) + 1 = 54$

No. of column completed : $3 - 47 - (47 - 3 + 1) = 45$

No. of element in each column $(100 - (-4) + 1) = 105$

total $= 45 \times 105$

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Topic : 2-D Column Major Order

Consider the following array A[-4…100][3…100]. Base address is 1000 and each element occupies 2-Bytes of space. What is the address of A[50][48] in column major order ?



Topic : 2-D Column Major Order

We need to find the address of A[50][48]. Before reaching to 48th column 3…47 columns i.e. 47-3+1=45 columns are already been arranged (Second way is (48- lower bound)) and each column consists of 105 elements. So total 45× 1005 =4725 elements are arranged.

For the 50th row we have already arranged -4…49 i.e. 54 (49 – (-4)+1) elements (look at the first dimension) are already arranged.

So total number of elements arranged are = 4725+55 = 4779.

The address is 1000+4779× 2=10558

$A[lb_1..ub_1][lb_2..ub_2]$

Row major order :-

Address of $A[i][j]$: $BA + [(i - lb_1) \times (ub_2 - lb_2 + 1) + (j - lb_2)] \times$ Size

Column major order :-

Address of $A[i][j] = BA + [(j - lb_2)(ub_1 - lb_1 + 1) + (i - lb_1)] \times Size$

Sparse Matrix

Sparse matrix are those matrix where relatively few entries are Non Zero.

to store them in optimize way we store only Non Zero entries.

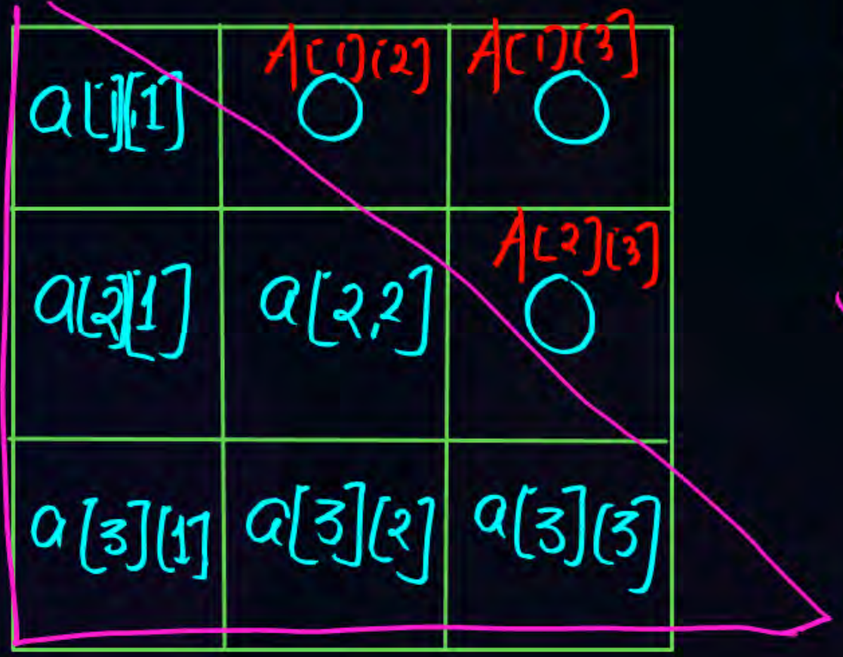


Topic: Lower Triagular Matrix

LTM (Square Matrix)

Rows = Columns

Lower Triangular matrix is a matrix in which element above principle diagonal are zero



what is the condition on [i,j]

Such that A[i][j] = 0

j > i A[i][j] = 0

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Topic: Lower Triagular Matrix

LTM (Square Matrix)

Rows = Columns

Lower Triangular matrix is a matrix in which element above principle diagonal are zero

a[1][1]	A[1][2] ○	A[1][3] ○
a[2][1]	a[2,2]	A[2][3] ○
a[3][1]	a[3][2]	a[3][3]

No. of Non zero element in LTM :

A[1..3][1..3] : No. of Non zero = 6

1+2+3 = 6

A[1..n][1..n] :

$$1+2+3+4+\cdots+n = \sum_{i=1}^{n} i = \frac{n(n+1)}{2}$$

Slide

Topic: Lower Triagular Matrix

* Row major order : only Non Zero element will be stored
* Column major order one after another

using formula

$$1000 + \left[\frac{3 \times 2}{2} + 1\right] \times 2$$

$1000 + 4 \times 2$

$= 1008$

a[1][1] ✓	A[1][2] ○	A[1][3] ○
a[2][1]	a[2,2]	A[2][3] ○
a[3][1]	a[3][2]	a[3][3]

Row major order :

BA : 1000

Size : 2B

a[1][1]	a[2][1]	a[2][2]	a[3][1]	a[3][2]	a[3][3]
1000	1002	1004	1006	1008	# 1010

Address of A[3][2] : if we are in 3rd Row

No. of Rows completed : 2

No. of element in 2 Rows : 1 + 2 = 3

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Topic: Lower Triagular Matrix

$A[1..n][1..n]$

$\frac{n(n+1)}{2} - \frac{(i-i)(i-1+i)}{2}$

Address of $A[i][j]$: if we are in i^{th} Row No. of Rows completed $= i-1$

No. of elements completed : $1+2+3+\cdots+(i-1) = \frac{i(i-1)}{2}$

if we are in j^{th} column No. of columns completed : $j-1$

Address $A[i][j] = BA + \left[\frac{((i-1)}{2} + (j-1)\right] \times Size$

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2 mins Summary

Topic

Topic

Topic

Topic

Topic

2-D array Column major order

Lower Triangular matrix

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THANK - YOU

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